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Human-centered Design Approach toward the Physical Activity Initiative for Work-from-Home Workers during the COVID-19 Outbreak

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Abstract: Physical activity levels have been declining since the COVID-19 outbreak due to social distancing and working from home. With respect to the modifiable risk factor of work-from-home (WFH) workers' physical inactivity, the present study seeks to develop a physical activity initiative for them during this global pandemic. Using the human-centered design (HCD) approach with the online involvement of twenty-two WFH workers, the results of this study presented the assessment and insights from the sequential HCD process regarding WFH workers' workstation in the home environment, views on received and tracked daily physical activities during the outbreak, and feedback from a five-day live physical activity prototyping. This study discusses the systematic development of the HCD approach in designing a physical activity initiative. However, a low number of participants may limit the conclusions drawn from this research. In conclusion, the HCD approach provided an opportunity to engage WFH workers and create and design with them to promote physical activity during a pandemic. After being developed through design development, being examined through hands-on use, and tested on the target group in pandemic settings, this present study informs and provides insights to design practitioners to consider the practical design of health workers during a pandemic.

Keywords: Covid-19 Outbreak, Health Initiative, Human-centered Design (HCD), Physical Activity, Work-From-Home (WFH)

Introduction

On January 30, 2020, the COVID-19 outbreak was declared by World Health Organization (WHO) as a Public Health Emergency of International Concern (WHO 2020a; Mahtani et al. 2020) and then proclaimed as a pandemic on March 11, 2020 (WHO 2020b). Globally, this outbreak has impacted people's activities, health, and well-being (UNDP 2020; Fore 2020). The coronavirus mainly spreads through close contact (Rothan and Byrareddy 2020; Sohrabi et al. 2020) via small droplets induced by sneezing, coughing, and talking (WHO 2020c; CDC 2020). People have been advised to practice social distancing (Wilder-Smith and Freedman 2020), leading government officials in many countries to regulate their citizens by ensuring they stay at home and practice physical distancing to fight the virus (Viner et al. 2020). Subsequently, similar to many countries, in Indonesia also, under government regulations, companies or institutions have been instructed to operate from home (DJalante et al. 2020; Ryalino 2020). Public- and private-company workers in Indonesia have started working remotely from their home, and thus, they are now called work-from-home (WFH) workers..

Although working from home reduces the possibility of being afflicted with the virus, this circumstance poses a new health problem. Unlike in developed countries, where companies help employees to set up a workstation at home (Harvard Business Review 2020; Nova 2020; Morel 2020), WFH workers in Indonesia receive little support from their employers. Some have a workstation with limited facilities while some do not even have a table (Umah 2020). The unfriendly working environment at home and limited access to the outdoor environment to

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avoid getting infected by the virus has, therefore, resulted in more sitting time inside the house and a lack of physical activity, leading to physical inactivity.

Physical (In)Activity

Physical activity is defined as any bodily movement produced by skeletal muscles, resulting in energy expenditure. In daily life, physical activity can be categorized into occupational, sports, conditioning, household, or other activities requiring a minimum of 150 minutes per week (Caspersen, Powell, and Christenson 1985; WHO 2010a). In contrast, physical inactivity is a term used to identify people who do not perform the WHO recommended level of regular physical activity (WHO 2010a, 53; The US New York Health n.d.).

While physical activity is associated with numerous health benefits, for both physical and mental health (Janssen and LeBlanc 2010), physical inactivity is a significant contributor to mortality. According to the WHO (Warburton et al. 2006; WHO 2010b), approximately 3.2 million deaths worldwide annually can be attributed to physical inactivity. It is a modifiable risk factor for cardiovascular disease and a widening variety of other chronic diseases, including diabetes mellitus, cancer (colon and breast), obesity, hypertension, bone and joint diseases (osteoporosis and osteoarthritis), and depression (WHO 2010a; WHO 2010b). Apparently, since the COVID-19 outbreak, physical activity levels have been declining (Hall et al. 2020; Jakobsson et al. 2020; Hammami et al. 2020) due to social distancing and working from home.

The Human-centered Design (HCD) in Health Research

Human-centered design (HCD) is an approach to developing a design solution by involving the human or potential users' perspectives and learning from them in all steps of the design process (Steen 2011). HCD "is based on the use of techniques that communicate, interact, empathize and stimulate the people involved, obtaining an understanding of their needs, desires, and experiences which often transcends that which the people themselves actually realized" (Giacomin 2014, 610). Scholars have examined the HCD in a variety of contexts and settings, specifically in the context of health. Previous research on the HCD has been used to develop an online intervention for the self-management of asthma (Morrison et al. 2015) and concepts to stimulate a vibrant local economy in the Oakland Best Babies Zone for infant mortality and community health (Vechakul, Shrimali, and Sandhu 2015). The HCD was explicitly used to understand the situation by collecting and analyzing data, developing a new clinical decision support system, and implementing and evaluating the system across twenty-four clinics in the study of Catalani et al. (2014). Furthermore, the HCD was used to design a backpack, which decreased the load on the shoulders, neck, and waist (Amiri, Dezfooli, and Mortezaei 2012), and global health education programs (Adam et al. 2019). Although the HCD has proved to be an effective approach to engage with stakeholders, users, and patients (Ragouzeos et al. 2018; Aifah et al. 2020), there is still a lack of research on HCD in the physical activity area. Thus, there exists an exciting opportunity to explore the HCD approach to promote physical activity among WFH workers during a pandemic.

Study Objective

This study aimed to develop a physical activity design initiative for WFH workers using the principles of the HCD in a pandemic setting. The initiative aims to incorporate better physical activity health literacy and support tracking activities into the daily physical activity routine of WFH workers. Through a sequential HCD process, this study described a comprehensive approach to use the HCD to generate physical activity design outcomes and initiative to inform

design and health professional practitioners, as well as contribute to the HCD research in pandemic settings.

Methods and Materials

A version modified by previous researchers of the IDEO approach to the HCD was applied in this study (IDEO 2015; Aifah et al. 2020). By using the principles of the HCD approach, this study aimed to develop a physical activity design initiative for WFH workers to stay active at home during a global pandemic. The core tenets of the HCD included in this study follow three sequential phases: inspiration, ideation, and implementation. Each phase involved key activities (Ragouzeos et al. 2018; Aifah et al. 2020). Empathizing and defining activities were included in the inspiration phase via interviews. Ideation covered design conceptualizing and rapid prototyping activities via co-creation. Implementation was incorporated with live prototyping and testing evaluation activities. The present study was conducted during the COVID-19 outbreak. Therefore, to abide by government order and regulation to practice physical/social distancing, data collection and all other activities were performed online.

Participants

Participants were selected through convenience sampling. They were sent emails, inquiring about their willingness to participate in this research study. Information about their characteristics, including information about their age and sex, was obtained. Participants were recruited during the first stage of study, and consent to participate in the research (brief demographic survey, online interviews, and workshop) was obtained through a written form. Participants were compensated for their participation. Participants provided consent through a written form prior to participation.

The HCD Process and Physical Activity Design Initiative Development

Phase I—Inspiration: Interview of WFH Workers

The inspiration phase involved empathizing and defining activities via online interviews (Zoom video meeting). Observation of participants' work environment was also conducted through the live video meeting. This phase aimed to empathize with and understand and define the challenges WFH workers experience in performing physical activity during a global pandemic. To do so, participants were asked to take a picture of their home workstation and show it. Furthermore, participants were asked about their physical activities, sitting time, and how they manage their physical activities daily during the outbreak, especially given the limitations to stepping outside the house. They were also questioned about their knowledge and recommendations about and the challenges they experience related to physical activity during the outbreak. The interview lasted approximately fifteen minutes and consisted of five main questions. Twelve WFH workers participated in the interview.

Phase II—Ideation: Co-creation with WFH Workers

For phase 2, participants were asked to attend an online co-creation workshop via a Zoom meeting. Co-creation is an opportunity to “convene a group of people you are designing for and then bring them into the design process” (IDEO 2015). Therefore, two main activities were performed as part of this phase: design conceptualization and ideation with rapid prototyping. While design conceptualization aims to enable participants to extend the practice of empathy to visioning, i.e., developing concrete solutions or ideas for design outcome (Ragouzeos et al. 2018), ideating with rapid paper prototyping is an effective way to make ideas and concepts tangible. Besides, rapid paper prototyping is an efficient design technique to learn and discuss through making for obtaining critical feedback from participants quickly. Co-creation was run

through a forty-minute online workshop to discuss physical activity health literacy resources, self-tracking, visual design, and thoughts on rapid paper prototyping. Eight WFH workers joined the ideation phase. All of them had participated in the previous phase (inspiration).

Phase III—Implementation: Live Prototyping and Testing

Phase III incorporated information from phases I and II into the iteration design of prototype design outcomes. Seventeen participants participated in a five-day prototype testing. Seven were involved in phases I and II. New participants (N = 10) were asked to partake in the live prototype testing. All participants were informed they would receive a series of mailings on physical health information during a five-day timeframe. They were asked to self-report and track their daily physical activity via journaling. On the final day, participants were asked to fill out a self-reported online questionnaire to assess and share their experience on the live prototype testing.

Data Collection and Analysis

Recordings, notes, Zoom screen captures, digital photographs, and prototypes generated from the interviews and focus groups were analyzed to establish a series of themes, insights, and reflections at the end of each phase based on preparation activities and questions in previous studies related to the HCD approach (Ragouzeos et al. 2018; Aifah et al. 2020). Additionally, this study included descriptive statistics used to describe the quantitative results of the implementation phase (Pocuca et al. 2015; Selamet and Simpson 2019). Bootstrapping was deployed to account for the small sample size of live prototyping (N = 17) and any potential skew. This synthesized data was used to establish goals related to the types of support, participant understanding, and experiences stemming from designing physical activity initiatives.

Results

Characteristics of WFH Workers

Table 1 presents the characteristics of the participating WFH workers. Female participants consistently dominated the study, with them comprising 66% of the participants in Phase I, 62.50% in Phase II, and 58.82% in Phase III. Most participants, i.e., over 85%, were employed full-time and were over 30 years old. Phase I consisted of participants with a mean height of 163.58 centimeters (SD = 7.06) and a mean weight of 69.09 kilograms (SD = 11.61), Phase II consisted of participants with a mean height of 165.62 centimeters (SD = 6.09) and a mean weight of 71.25 kilograms (SD = 13.51). Phase III included participants with a mean height of 164.35 centimeters and a mean weight of 68.29 kilograms (SD = 11.42).

Table 1: Characteristics of WFH Workers

<i>Characteristics</i>	<i>Phase I (Inspiration) N = 12</i>	<i>Phase II (Ideation) N = 8</i>	<i>Phase II (Implementation) N = 17</i>
Age, mean (SD)	33.50 (3.52)	32.75 (3.99)	33.17 (5.28)
Female, n (%)	8 (66.66%)	5 (62.50)	10 (58.82%)
Employment, Fulltime, n (%)	11 (91.66%)	7 (87.50)	15 (88.23%)
Height, mean (SD)	163.58 (7.06)	165.62 (6.09)	164.35 (6.65)
Weight, mean (SD)	69.08 (11.61)	71.25 (13.51)	68.29 (11.42)

Source: Selamet

Insights from Inspiration: Defining Problems and Challenges

Table 2 presents the insight themes, description, frequency, and proposed action from Phase I. Three primary insights were generated from the inspiration phase: working from home during a global pandemic should consider the home condition of WFH workers, WFH workers need health literacy resources, and they need motivation for physical activity.

Table 2: Insights from Inspiration: Defining Problems and Challenges

<i>Thematic Insights</i>	<i>Description</i>	<i>Frequency</i>	<i>Proposed Action</i>
Global pandemic impacting WFH workers	Government regulations for stay-at-home order	General	Provide an alternative solution to maintain health
	Limited outside activities	General	
	Working from home without personal preparation	General	Provide a series of mails on information to be active in a limited working environment
	Adjusting without office activities	Typical	
WFH workers need resources on health literacy	Need for information on physical activity during a pandemic	Typical	Promote easy access to information on physical activity
	Information on physical activity requires to be credible and standardized	General	
Motivation for performing physical activity	WFH workers value personal motivation to perform physical activity; Need for personalized motivation to perform physical activity	Typical	Provide an engaged physical activity design
	Tracking physical activity may motivate WFH workers	Typical	

Source: Selamet

Note. *General*—core ideas appealed to all participants; *typical*—core ideas of the category appealed to five to eight participants; *variant*—core ideas of the category appealed to two to four participants.

Global Pandemic and Condition of WFH Workers

In general, participants acknowledged the government's stay-at-home orders, limiting outside activities. Moreover, the order impacted their employers, who instructed workers to work from home. However, participants, in general, were not well prepared to follow this command. Moreover, their employers did not provide any support to set up workplace stations at home. Participant IN-K said, "I live in four-by-four meters square room; I do not have a worktable. I work on my bed sadly." Participant IN-U stated, "I used to walk around my office during the break or take a walk outside the office, like a 15-minute walk; [I] cannot do that now. I sit more since working at home, and I am getting fat." Participant IN-H also stated the same: "this sudden adjustment and transformation that I have not experienced in my ten years of career. At the office, I used to take stairs or go to my friend's desk to talk; now, with this online meeting, it all sits, sit and sit; I did not have experience working at home. With this condition, I can tell I sit more than before." Participants voluntarily sent pictures of their workstation at home to illustrate their working conditions (see Figure 1).



Figure 1: Some of Photo Collections of Participant's Workstations at Home
Source: Selamat

Health Resources on Physical Activity during a Pandemic

All participants received emails from their employers on basic health practices one should adhere to during a pandemic, such as washing hands and wearing a mask. Some even received a tracking app from their company to track their health so they could directly report to the company whether they have coronavirus. However, none of them received any information regarding physical activity health resources while performing WFH activities. Participants felt those resources would have helped advocate and remind them to stay active and healthy. Participant IN-J said, "it may seem small details, but the employee needs a health literacy resources; small tips for WFH workers would help." IN-L expressed the same desire: "physically and mentally, I am not all prepared for this WFH situation, so small advocacy what to do during this WFH moment would be great." Participant IN-E stated, "[I] received nothing from my employment about how to stay active during this pandemic. So, I look up online literacy. Then I overwhelmed with all the information. I concern about the credibility of some

information that I have found online. It would be great if I could receive health literacy from a well-known institution. I need credible health information.”

Motivation for Physical Activity

In general, participants needed the motivation to stay active at home. They were concerned about the impact of physical inactivity on their health and well-being. Typically, participants acknowledged their limitations such as space but were willing to overcome the situation and work on their health. “A reminder with some health info, I think to me, may encourage me to stay active at home,” stated participant IN-A. Participant IN-H vocalized, “regular daily health literacy information and tracking activity that I can control or self-management that would help me to stay active.” Participant IN-G said, “I honestly lost track sometimes, like times and days, you know when you stay at home alone for too long. So, I don’t know if I am being active or not.” Participants thought tracking physical activity would help motivate them to stay active, while simultaneously arguing they needed practical information to guide them to do a physical activity while working from home.

Design Insights from the Ideation Phase

Three main insights emerged from Phase II, the ideation activity, as seen in Table 3. In the ideation phase, co-creation with participants was conducted to conceptualize and develop the design process based on the understanding gained in Phase I, i.e., the inspiration phase. From this activity, these insights were outlined to provide physical activity information, design physical activity communication materials, and track physical activity.

Table 3: Design Insights from Ideation; Conceptualization and Design Development with WFH Workers

<i>Thematic Insights</i>	<i>Representative Quote(s)/Activities/Features</i>	<i>Frequency</i>	<i>Proposed Action</i>
WFH workers suggest clear and/or appropriate physical activity information content	Acceptable, interesting, and entertaining	Typical	Adopt physical activities’ framework from WHO for credible information
	Easy to understand	General	
	Call for action	General	
	Engaging content	Typical	
	About 5–7 minutes to read	Typical	
On designing physical activity communication materials	Get visuals: Ability to illustrate, images of inspiring objects, people, sites, or pets	Typical	Conduct co-session design with WFH workers to discuss the visual design
	Easy-to-read visual messages	Typical	
	Well-conceptualized	Variant	
On tracking physical activity	Personalized physical activity tracking	General	Provide digital a physical-activity journal diary that can be printed or typed online
	Online physical activity tracking	Variant	
	Options for both writing and typing to track physical activity	Typical	

Source: Selamet

Note. *General*—core ideas appealed to all participants; *typical*—core ideas of the category appealed to five to six participants; *variant*—core ideas of the category appealed to two to four participants.

Need for Frameworks for Physical Activity Information

While many participants recognized that information on physical activity resources could be easily found online, they, however, intended to acquire appropriate information on physical activity content from credible sources targeted to them. In general, participants felt comfortable following physical activity guidelines from highly credible resources. Participant ID-A suggested “a clear framework guidelines would benefit my physical activity behavior, and please make sure that physical activity guidelines not for a targeted athlete so a person like me can do it.” Therefore, to facilitate this need, WHO frameworks and guidelines were presented.

On Designing and Delivering Physical Activity Communication Materials

In general, participants rejected all text content with no visuals and recommended using more visuals. During the online co-session, participant ID-R said, “visual information helps me to stay engaged.” Participants discussed how visual messages should be clear, simple, and easy to read. Moreover, they expressed the fear of not being able to engage with the extended information and suggested having information up to five minutes of reading. Paper-based prototypes were presented for discussion to iterate and facilitate the co-creation meeting.

All co-creation participants said they had prepared to receive health literacy resources on physical activity through their work email. Participant ID-K stated, “I am sitting more during this [working-from-home] time, especially in the weekdays; a piece of physical activity information deliver through my work email will remind me to, I do not know, stand up and walk around my room, maybe.” Agreeing with ID-K, participant ID-H said, “yes, work email would be great to deliver this information. I check my work email more often during this pandemic, maybe because I mostly spend my day at home looking to my computer screen.”

On Tracking Physical Activity

Co-creation started with a discussion on the framework and design of the physical activity and its information content, respectively. The final part of the discussion pertained to tracking physical activity. While participants recognized that literacy information on physical activity via email might encourage and remind them to perform physical activity, they wanted to put their knowledge from the content into action to track and manage it. Tracking physical activity emerged from the discussion. Although various participants wanted to have a digital app, they did not want to spend more money on purchasing it. Further, participants outlined the need for physical activity tracking that could be easily personalized. A physical-activity-tracking journal was presented. From the series of sketch paper prototypes, participants suggested being provided with the option to write in their journal online or offline. In other words, they wanted to have the option to both write and type to track their physical activity. While participant ID-K wanted a feature that could summarize the duration of the physical activity, participant ID-R added, “I want my journal to have space for writing notes. I like notes.”

Insights from Implementation

Live Prototyping. Participants (N=17) received daily physical activity information for five days that sent to their email (see Figure 3). In addition to receiving daily physical health information literacy, physical activity tracking journal (Figure 2) was sent to participants to report their physical activity daily. Insight results from the live prototyping was reported in Table 4, 5, 6 and 7.

Table 4: Perceived by Receiving Health Literacy Information on Physical Activity during the COVID-19 Outbreak (self-reported) (N = 17)

Item	M (SD)	95% CI (a)	
		LL	UL
I liked receiving the information sent to me.	3.94 (1.08)	3.38	4.50
I liked the wording of the information sent to me.	3.88 (1.16)	3.28	4.48
The health information is useful for me.	4.29 (0.68)	3.94	4.64
I engaged in learning from the information sent to me.	3.47 (0.87)	3.02	3.92
The information helped me understand the importance of physical activity	4.05 (0.65)	3.72	4.39
The information helped me understand the negative effects of sedentary behavior	4.00 (0.61)	3.68	4.31
The information gave me helpful tips on how to be active at home during the COVID-19 outbreak	4.05 (0.96)	3.56	4.55
The information motivated me to be active at home during the COVID-19 outbreak	3.58 (1.00)	3.07	4.10

Source: Selamet

Note. CI = confidence interval; LL = lower limit, UL = upper limit.

Variables were scored on a scale from 1 to 5, i.e., from strongly disagree to strongly agree (1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree).

(a) Confidence intervals are bootstrapped to account for potential skew in a small sample size.

My Physical Activity Journal
Stay Active > 30 minutes of Physical Activity a Day - Prototype

MONDAY				TUESDAY				WEDNESDAY				THURSDAY				FRIDAY				
No	Time of Day	Description of Activity	Duration (minutes)	Note	Time of Day	Description of Activity	Duration (minutes)	Note	Time of Day	Description of Activity	Duration (minutes)	Note	Time of Day	Description of Activity	Duration (minutes)	Note	Time of Day	Description of Activity	Duration (minutes)	Note
Total				Total				Total				Total				Total				

OVERALL NOTES

TO DO

Figure 2: The Live Prototyping-Physical Activity Journal

Source: Selamet



Figure 3: The Live Prototyping Health Literacy Sent via Email for Five Days; Adapted From WHO Physical Activity Guidelines and Move Your Way Campaign
Source: Selamet

Receiving Health Literacy Information on Physical Activity

Overall, Table 4 shows that the participants liked receiving health literacy information on physical activity during the COVID-19 outbreak. Participants scored the following variables a four: usefulness ($M = 4.29$, $SD = 0.68$) of understanding the importance of physical activity ($M = 4.05$, $SD = 0.65$), usefulness of understanding the negative effects of sedentary behavior ($M = 4.00$, $SD = 0.61$), and helpfulness of the tips on how to be active during the COVID-19 outbreak ($M = 4.05$, $SD = 0.96$). Participants scored the following a three: they feel okay about receiving the information ($M = 3.94$, $SD = 1.08$), the information's wording ($M = 3.88$, $SD = 1.16$), and motivation ($M = 3.58$, $SD = 1.00$). Participants gave the lowest score to the engagement of the content ($M = 3.47$, $SD = 0.87$).

Perception of Writing a Journal to Track Physical Activity Daily

Table 5 shows that the physical-activity-tracking journal helped participants track their physical activity ($M = 3.70$, $SD = 0.98$) and made them more active during the outbreak ($M = 3.88$, $SD = 1.11$). After tracking their physical activity for five days, participants felt proud of their physical activity achievement ($M = 3.70$, $SD = 1.10$). Working on their physical-activity tracking journal made them feel it might influence their physical activity behavior to be more active at home during the COVID-19 outbreak ($M = 4.00$, $SD = 0.86$).

Table 5: Perception of Writing a Journal to Track Physical Activity Daily for Five Days ($N = 17$)

Item	$M (SD)$	95% CI (a)	
		LL	UL
Writing in the physical-activity-tracking diary made me more active during the outbreak.	3.88 (1.11)	3.31	4.45
Writing in the physical-activity-tracking diary helped me track my physical activity behavior.	3.70 (0.98)	3.19	3.21
After writing in my journal, I was proud of my physical activity achievement.	3.70 (1.10)	3.13	4.29
The writing may influence my physical activity behavior to be more active at home during the COVID-19 outbreak	4.00 (0.86)	3.55	4.44

Source: Selamet

Note. CI = confidence interval; LL = lower limit, UL = upper limit.

Variables were scored on a scale from 1 to 5, i.e., from strongly disagree to strongly agree (1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree).

(a) Confidence intervals are bootstrapped to account for potential skew in a small sample size.

Overall Design of the Physical Activity Prototype Initiative Evaluation Scores

Table 6 outlines the overall score assigned to the design prototype by those who participated in prototype testing. Overall, the prototype received a good score, ranging from the lowest score where participants could not decide whether the quantity information was too much or about right ($M = 3.00$, $SD = 1.41$) to the highest score where they would recommend that the prototype reach more people ($M = 4.23$, $SD = 0.97$). The overall star rating of the prototype has a mean of 3.94 ($SD = 0.65$) with a fair score in terms of visual design ($M = 3.64$, $SD = 1.11$) and aesthetics aspect ($M = 3.47$, $SD = 1.06$). The prototype has a good score ($M = 4.00$, $SD = 0.79$) in terms of the quality of information sent to the participants.

Table 6: Overall Experience in (with) Using/Getting Testing Prototype by Receiving Information and Writing a Physical-Activity Journal (N = 17)

Item	M (SD)	95% CI (a)	
		LL	UL
Aesthetics of the initial prototype design	3.47 (1.06)	2.92	4.01
Visual design/information of the prototype	3.64 (1.11)	3.07	4.22
Quality of prototype information	4.00 (0.79)	3.59	4.40
Number (quantity) of information of the prototype	3.00 (1.41)	2.27	3.72
Overall star rating of the prototype	3.94 (0.65)	3.60	4.27
Prototype recommendation to reach more people*	4.23 (0.97)	3.73	4.73

Source: Selamet

Note. CI = confidence interval; LL = lower limit, UL = upper limit.

(a) Confidence Intervals are bootstrapped to account for potential skew in a small sample size.

Variables are scored on a scale from 1 to 5, i.e., from bad to excellent (1 = bad, 2 = poor, 3 = fair, 4 = good, 5 = excellent). *1 = definitely not recommended, 2 = not recommended, 3 = undecided, 4 = recommended, 5 = highly recommended

Implementation of Qualitative Thematic Insights in Prototype Testing and Evaluation

Table 7 reveals two main themes that emerged. On receiving health literacy information via email, while asking to follow-up with questions about their experience on receiving the information, echoing their questioner's answers, participants consistently praised the prototype in terms of its ability to add to their knowledge on health literacy and facilitate physical information with well-crafted visual messages. However, some stated they that they left some of the emails unread, as they felt they were receiving too many. Participant IM-J said, "it was just like receiving a series of marketing emails." Furthermore, some were concerned about the language used. Participant IM-E said, "I understand English, but honestly, I think I would prefer to get the information in Indonesian.

Table 7: Qualitative Thematic Insights from Live Prototyping

Thematic Insights	Category	Frequency
On receiving health information via email	Adding to health literacy knowledge	Typical
	Unread email	Variant
	Facilitated physical activity information	Typical
	Appreciating well-crafted visual messages	Typical
	Prepared in Indonesian (not English) language	Variant
On reporting in a physical-activity journal	Self-reported more than 150 minutes of physical activity	Variant
	Self-reported about 150 minutes of physical activity	Typical
	House cleaning activity	General
	Dancing	Variant
	Walking around the house	General
	Yoga	Typical
	Home workout	Variant
	Time-consuming or takes time to write	Variant

Source: Selamet, 2020

Note. General—core ideas appealed to all participants; typical—core ideas of the category appealed to six to ten participants; variant—core ideas of the category appealed to two to five participants.

The second theme centered on reporting in the physical-activity journal. Various participants self-reported having performed more than 150 minutes of physical activity in five

days, whereas typically, participants reported having performed approximately 150 minutes of physical activity during the timeframe. All participants reported cleaning the house and walking around the house as their form of physical activity. Moreover, participants reported they danced, did yoga, and worked out at home during their stay at home. Most participants reported benefiting from the activity journal, as quoted by participant IM-S, “writing a physical-activity journal is just like self-evaluation for me, whether I am an active person or not. Therefore, this motivates me to become more active because I want to be an active person.” Nevertheless, some participants felt that writing a physical-activity journal is time-consuming and ineffective, as quoted by participant IM-J, “sometimes, I forget my physical activity. I cannot just go for a morning run and back home to write a journal. It is not effective and time-consuming. Digital tracking is way easier than this [journal].”

Discussion

The prototype of the physical activity design initiative for WFH workers was developed using an iterative HCD process. This initiative aimed to support and motivate WFH workers during the COVID-19 outbreak. During the development process, this study carefully followed the HCD approach to assess and consider WFH workers’ views and participation through inspiration, ideation, and implementation. Therefore, this study led to an initiative adaptation that was WFH-worker-driven (participants), with testing evaluation presented.

Results from the HCD process informed the uniqueness of this physical activity design initiative in several ways. First, the inspiration phase/stage provided the opportunity to learn and understand WFH workers’ current work conditions during a global pandemic. It informed and recognized their condition, and their interests, motivation, and views on how they can be best supported in their physical health activity were identified. Specifically, it was only through this process that WFH workers’ work environment at home could be observed online and learned. Results and insights from this stage became the preliminary recommendation for the ideation phase.

Second, the literature suggested that the HCD promises to facilitate deep and meaningful involvement of participants in developing the design and testing the prototype (Giacomin 2014; Ragouzeos et al. 2018; Aifah et al. 2020; Tjandra, Pinasthika, and Winantyo 2020). Therefore, informed by doing so in Phase I, Phase II (ideation) was conducted to accommodate the involvement of participants in design conceptualization and development through co-creation. The ideation phase provided a critical aspect to understand that information on physical activity needs to uncover clear physical activity guidelines that WFH workers can follow. Additionally, this study discovered the participants’ desire to put their knowledge into action through this process. While conceptualizing physical information literacy content and delivery scenario, suggestions to put health literacy into action were developed into tracking physical activity. Insights from paper-based prototype testing design were useful in discussing design development. Without ideation, design direction might have easily gone unnoticed, and participants would have been unlikely to access and use the design outcome, had they found it interesting, and reflect on and represent their current condition.

Third, by performing the implementation phase, this study presents live prototyping and testing evaluation. Phases I and II were informed about prototype design development before it went to testing. Evaluation of live prototype testing revealed that a critically perceived benefit of the initiative was the importance of physical health literacy and physical activity tracking to support WFH workers’ needs during a pandemic. Findings of the live prototype testing yield valuable feedback regarding WFH workers’ communication design and tracking system preferences. Together, results from the live prototype testing highlight that, in general, WFH workers showed keen interest in and support for the initiative aimed to help them stay active during the outbreak.

Finally, this study yields the HCD process in approaching participatory engagement from the beginning, mainly as it ensures outcomes are driven by unique preferences, feedback, and circumstances of the potential target group. Similar to previous studies (Ceschin and Gaziulusoy 2016; Ragouzeos et al. 2018; Aifah et al. 2020), it shows how the HCD offers the opportunity to define, create, and develop WFH- worker-driven solutions based on their current condition. Through the HCD, this study presents an effort to test and evaluate live prototyping to learn how this physical activity initiative is perceived by WFH workers.

Limitations

This study had limitations. The number of participants in the inspiration ($N = 10$), ideation ($N = 8$), and live prototype testing evaluation phases ($N = 17$) were small, raising the question whether a larger sample group of WFH workers could have led to more feedback, insights, and opportunities for rigorous design outcomes. There is a debate among the HCD researchers regarding the most cost-efficient number of participants in an HCD process (Selamet and Simpson 2019). Vilardaga et al. (2018) argued that the most traditional approach suggested a sample size of five. However, recruiting WFH workers during the outbreak is particularly challenging, and thus, the author believes this study's sample size was justified.

The HCD process could have been applied more rigorously by including all key stakeholders in the study. As reported by previous studies (Aifah et al. 2020), participation from key stakeholders of different backgrounds and expertise may be impacted by intervention, initiative, or program. Future studies should put effort into including all key stakeholders in an HCD process. Additionally, a short five-day period of live prototyping without a control group may limit the conclusions drawn from the present study. Future work that fully implements the HCD process should include an evaluation of the design initiative's health outcomes, as suggested by Bazzona et al. (2017).

Conclusion

In summary, principles of the HCD approach were systematically used to develop a prototype of a physical activity initiative that supports WFH workers to stay active at home during the challenging times of a global pandemic with limitations on going outside. The use of this iterative HCD approach provided an opportunity to engage WFH workers and create and design with them to understand their current work environment and situation. More importantly, the current study presented live testing and evaluation of the prototype and generated feedback, which can be used to enhance the literature for future studies related to the HCD.

Moreover, after being developed through design development, being examined through hands-on use, and being tested on the target group in a pandemic setting, this present study may inform and provide insights to design practitioners working on the effective design about the health of workers during a pandemic.

REFERENCES

- Adam, Maya, Shannon A. McMahon, Charles Prober, and Till Bärnighausen. 2019. "Human-Centered Design of Video-Based Health Education: An Iterative, Collaborative, Community-Based Approach." *Journal of Medical Internet Research* 21 (1): e12128. <https://doi.org/10.2196/12128>.
- Aifah, Angela, Nwora Lance Okeke, Cynthia R. Rentrop, Julie Schexnayder, Gerald S. Bloomfield, Hayden Bosworth, Kiran Grover, et al. 2020. "Use of a Human-Centered Design Approach to Adapt a Nurse-Led Cardiovascular Disease Prevention Intervention in HIV Clinics." *Progress in Cardiovascular Diseases* 63 (2): 92–100. <https://doi.org/10.1016/j.pcad.2020.02.013>.
- Amiri, Mahdie, Mohsen Safar Dezfooli, and Seyed Reza Mortezaei. 2012. "Designing an Ergonomics Backpack for Student Aged 7-9 with User Centred Design Approach." *Work* (Reading, Mass.) 41 Suppl 1: 1193–1201. <https://doi.org/10.3233/WOR-2012-0303-1193>.
- Caspersen, C. J., K. E. Powell, and G. M. Christenson. 1985. "Physical Activity, Exercise, and Physical Fitness: Definitions and Distinctions for Health-Related Research." *Public Health Reports* 100 (2): 126–31. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1424733/>.
- Catalani, Caricia, Eric Green, Philip Owiti, Aggrey Keny, Lameck Diero, Ada Yeung, Dennis Israelski, and Paul Biondich. 2014. "A Clinical Decision Support System for Integrating Tuberculosis and HIV Care in Kenya: A Human-Centered Design Approach." *PLOS ONE* 9 (8): e103205. <https://doi.org/10.1371/journal.pone.0103205>.
- Centers for Disease Control (CDC). 2020. "Coronavirus Disease 2019 (COVID-19) - Transmission." *Centers for Disease Control and Prevention*, June 1, 2020. <https://www.cdc.gov/coronavirus/2019-ncov/prevent-getting-sick/how-covid-spreads.html>.
- Ceschin, Fabrizio, and Idil Gaziulusoy. 2016. "Evolution of Design for Sustainability: From Product Design to Design for System Innovations and Transitions." *Design Studies* 47 (November): 118–63. <https://doi.org/10.1016/j.destud.2016.09.002>.
- Djalante, Riyanti, Jonatan Lassa, Davin Setiamarga, Aruminingsih Sudjatma, Mochamad Indrawan, Budi Haryanto, Choirul Mahfud, et al. 2020. "Review and Analysis of Current Responses to COVID-19 in Indonesia: Period of January to March 2020." *Progress in Disaster Science* 6 (April): 100091. <https://doi.org/10.1016/j.pdisas.2020.100091>.
- Fore, Henrietta H. 2020. "A Wake-up Call: COVID-19 and Its Impact on Children's Health and Wellbeing." *The Lancet* 8 (7): E861–E862. [https://doi.org/10.1016/S2214-109X\(20\)30238-2](https://doi.org/10.1016/S2214-109X(20)30238-2).
- Giacomin, Joseph. 2014. "What Is Human Centred Design?" *The Design Journal* 17 (4): 606–23. <https://doi.org/10.2752/175630614X14056185480186>.
- Hall, Grenita, Deepika R. Laddu, Shane A. Phillips, Carl J. Lavie, and Ross Arena. 2020. "A Tale of Two Pandemics: How Will COVID-19 and Global Trends in Physical Inactivity and Sedentary Behavior Affect One Another?" *Progress in Cardiovascular Diseases*, April. <https://doi.org/10.1016/j.pcad.2020.04.005>.
- Hammami, Amri, Basma Harrabi, Magni Mohr, and Peter Krstrup. 2020. "Physical Activity and Coronavirus Disease 2019 (COVID-19): Specific Recommendations for Home-Based Physical Training." *Managing Sport and Leisure* 0 (0): 1–6. <https://doi.org/10.1080/23750472.2020.1757494>.
- Harvard Business Review. 2020. "Are You Suddenly Working from Home?" *Harvard Business Review*, April 2, 2020. <https://hbr.org/podcast/2020/04/working-while-distancing>.

- IDEO. 2015. "The Field Guide to Human-Centered Design." *IDEO.org*. <https://www.designkit.org/resources/1>.
- Jakobsson, Johan, Christer Malm, Maria Furberg, Ulf Ekelund, and Michael Svensson. 2020. "Physical Activity during the Coronavirus (COVID-19) Pandemic: Prevention of a Decline in Metabolic and Immunological Functions." *Frontiers in Sports and Active Living* 2. <https://doi.org/10.3389/fspor.2020.00057>.
- Mahtani, Shibani, Miriam Berger, Siobhan O'Grady, and Marisa Iati. 2020. "Hundreds of Evacuees to Be Held on Bases in California; Hong Kong and Taiwan Restrict Travel from Mainland China - The Washington Post." *Washington Post*, February 5, 2020. https://www.washingtonpost.com/world/asia_pacific/coronavirus-china-live-updates/2020/02/05/114ced8a-479c-11ea-bc78-8a18f7afcee7_story.html.
- Morel, David. 2020. "Is Home The New Office?" *Forbes*, May 26, 2020. <https://www.forbes.com/sites/davidmorel/2020/05/26/is-home-the-new-office/>.
- Pocuca, Nina, Leanne Hides, Oksana Zelenko, Lake-Hui Quek, Stoyan Stoyanov, Kristen Tulloch, Daniel Johnson, Dian Tjondronegoro, and David J. Kavanagh. 2016. "Initial Prototype Testing of Ray's Night Out: A New Mobile App Targeting Risky Drinking in Young People." *Computers in Human Behavior* 54 (January): 207–14. <https://doi.org/10.1016/j.chb.2015.07.033>.
- Ragouzeos, Dana, Julie Gandrup, Beth Berrean, Jing Li, Marie Murphy, Laura Trupin, Jinoos Yazdany, and Gabriela Schmajuk. 2019. "'Am I OK?' Using Human Centered Design to Empower Rheumatoid Arthritis Patients through Patient Reported Outcomes." *Patient Education and Counseling* 102 (3): 503–10. <https://doi.org/10.1016/j.pec.2018.10.016>.
- Rothan, Hussin A., and Siddappa N. Byrareddy. 2020. "The Epidemiology and Pathogenesis of Coronavirus Disease (COVID-19) Outbreak." *Journal of Autoimmunity* 109: 102433. <https://doi.org/10.1016/j.jaut.2020.102433>.
- Ryalino, Christopher. 2020. "How Indonesia Copes with Coronavirus Disease 2019 so Far (Part One): The Country, the Government, and the Society." *Bali Journal of Anesthesiology* 4 (2): 33. https://doi.org/10.4103/BJOA.BJOA_34_20.
- Selamet, Juhri, and Thomas Simpson. 2019. "A Design for Recycling: Developing a Mobile Application to Improve Recycling Behavior." *The International Journal of Visual Design* 13 (4): 45–61. <https://doi.org/10.18848/2325-1581/CGP/v13i04/45-61>.
- Sohrabi, Catrin, Zaid Alsafi, Niamh O'Neill, Mehdi Khan, Ahmed Kerwan, Ahmed Al-Jabir, Christos Iosifidis, and Riaz Agha. 2020. "World Health Organization Declares Global Emergency: A Review of the 2019 Novel Coronavirus (COVID-19)." *International Journal of Surgery* (London, England) 76 (April): 71–76. <https://doi.org/10.1016/j.ijsu.2020.02.034>.
- Steen, Marc. 2011. "Tensions in Human-Centred Design." *CoDesign* 7 (1): 45–60. <https://doi.org/10.1080/15710882.2011.563314>.
- Tjandra, Agatha Maisie, Lalitya Talitha Pinasthika, and Rangga Winantyo. 2020. "Developing Hybrid Media Installation Game in Taman Gajah Tunggal." *Ultimart: Jurnal Komunikasi Visual* [Ultimart: Journal of Visual Communication] 13 (1): 16–23. <https://doi.org/10.31937/ultimart.v13i1.1558>.
- The US New York Health. n.d. "Physical Inactivity and Cardiovascular Disease." Accessed June 3, 2020. <https://www.health.ny.gov/diseases/chronic/cvd.htm>.
- Umah, Anisatul. 2020. "Usai Pandemi Covid-19, Jam Kerja 9 to 5 Tinggal Kenangan? [After Pandemic Covid-19, are 9 to 5 working hours left to remember?]" *Lifestyle*. Accessed June 3, 2020. <https://www.cnbcindonesia.com/lifestyle/20200503201613-33-155918/usai-pandemi-covid-19-jam-kerja-9-to-5-tinggal-kenangan>.

- United Nations Development Programme (UNDP). 2020. “COVID-19 Pandemic: Humanity Needs Leadership and Solidarity to Defeat the Coronavirus.” Accessed June 3, 2020. <https://www.undp.org/content/undp/en/home/coronavirus.html>.
- Vechakul, Jessica, Bina Patel Shrimali, and Jaspal S. Sandhu. 2015. “Human-Centered Design as an Approach for Place-Based Innovation in Public Health: A Case Study from Oakland, California.” *Maternal and Child Health Journal* 19 (12): 2552–2559. <https://doi.org/10.1007/s10995-015-1787-x>.
- Vilardaga, Roger, Javier Rizo, Emily Zeng, Julie A. Kientz, Richard Ries, Chad Otis, and Kayla Hernandez. 2018. “User-Centered Design of Learn to Quit, a Smoking Cessation Smartphone App for People With Serious Mental Illness.” *JMIR Serious Games* 6 (1): e2. <https://doi.org/10.2196/games.8881>.
- Viner, Russell M., Simon J. Russell, Helen Croker, Jessica Packer, Joseph Ward, Claire Stansfield, Oliver Mytton, Chris Bonell, and Robert Booy. 2020. “School Closure and Management Practices during Coronavirus Outbreaks Including COVID-19: A Rapid Systematic Review.” *The Lancet Child & Adolescent Health* 4 (5): 397–404. [https://doi.org/10.1016/S2352-4642\(20\)30095-X](https://doi.org/10.1016/S2352-4642(20)30095-X).
- World Health Organization (WHO). 2010a. “WHO | Global Recommendations on Physical Activity for Health.” <https://www.who.int/dietphysicalactivity/publications/9789241599979/en/>.
- WHO. 2010b. “WHO | Physical Inactivity: A Global Public Health Problem.” Accessed May 3, 2020. https://www.who.int/dietphysicalactivity/factsheet_inactivity/en/.
- World Health Organization (WHO). 2020a. “Statement on the Second Meeting of the International Health Regulations (2005) Emergency Committee Regarding the Outbreak of Novel Coronavirus (2019-NCoV).” World Health Organization, January 30, 2020. [https://www.who.int/news-room/detail/30-01-2020-statement-on-the-second-meeting-of-the-international-health-regulations-\(2005\)-emergency-committee-regarding-the-outbreak-of-novel-coronavirus-\(2019-ncov\)](https://www.who.int/news-room/detail/30-01-2020-statement-on-the-second-meeting-of-the-international-health-regulations-(2005)-emergency-committee-regarding-the-outbreak-of-novel-coronavirus-(2019-ncov)).
- WHO. 2020b. “WHO Director-General’s Opening Remarks at the Media Briefing on COVID-19 - 11 March 2020.” World Health Organization, March 11, 2020. <https://www.who.int/dg/speeches/detail/who-director-general-s-opening-remarks-at-the-media-briefing-on-covid-19---11-march-2020>.
- WHO. 2020c. “Q&A on Coronaviruses (COVID-19).” World Health Organization, April 17, 2020. <https://www.who.int/news-room/q-a-detail/q-a-coronaviruses>.
- Wilder-Smith, A., and D. O. Freedman. 2020. “Isolation, Quarantine, Social Distancing and Community Containment: Pivotal Role for Old-Style Public Health Measures in the Novel Coronavirus (2019-NCoV) Outbreak.” *Journal of Travel Medicine* 27 (2). <https://doi.org/10.1093/jtm/taaa020>.